

TrES-3b

R-0440

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-3_180702 · created 2026-07-19T13:55:19+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458301.7065 +/- 0.0014 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.126 +/- 0.05
Transit depth (Rp/Rs)^2	1.59 +/- 1.25 [%]
Orbital Inclination (inc)	81.27 +/- 2.88
Airmass coefficient 1 (a1)	1.136 +/- 0.015
Airmass coefficient 2 (a2)	-0.12 +/- 0.023
Scatter in the residuals of the lightcurve fit is	1.28 %
Best Comparison Star	None
Optimal Aperture	3.71
Optimal Annulus	13.11
Transit Duration (day)	0.037 +/- 0.028

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.126 $\pm$ 0.05 vs 0.16309 (deviation 0.48 σ)
Duration fit vs published	0.037 d vs 0.0567 d (deviation 0.46 σ)
Mid-time O-C	2.7 min
Depth SNR	1.6
Residual scatter	1.28 %

Light curve

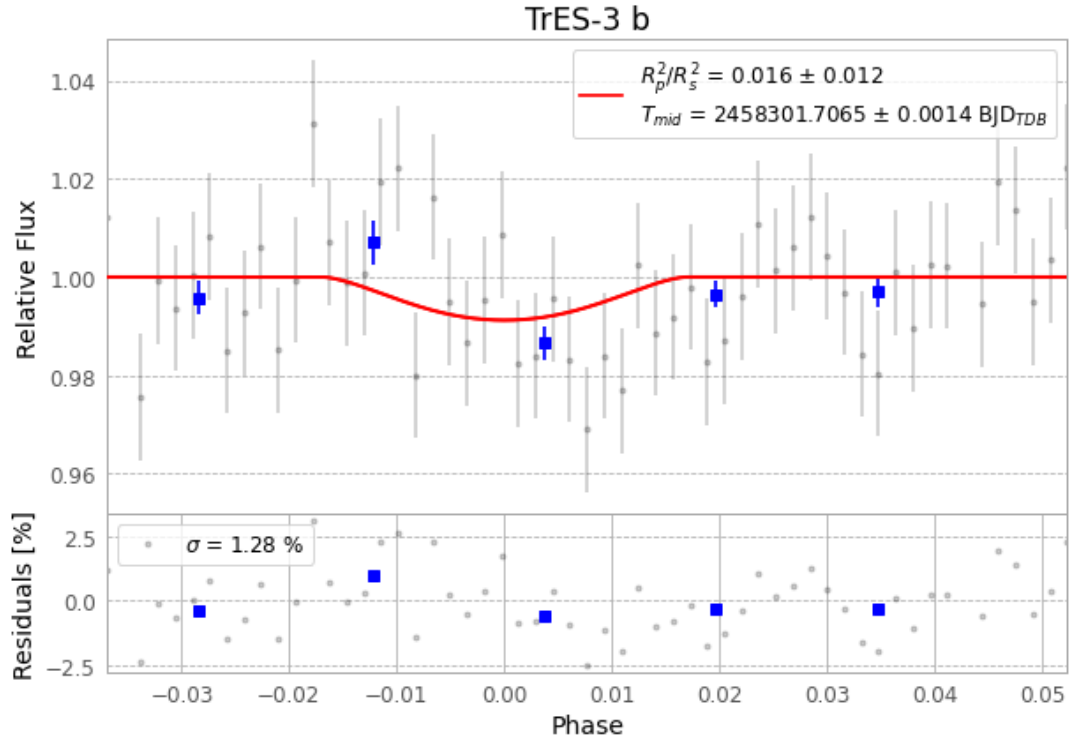


Figure 1 — FinalLightCurve_TrES-3 b_2018-07-01.png (SHA-256 d300032c23fc9d3d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [383, 206], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 4c75623b0200ba9d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 39d9b80

Data provenance

56 FITS frames (37 MB), TRES-3180702034215.FITS → TRES-3180702063012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-3-180702.log (e6aa5c0974e9...), FinalLightCurve_TrES-3 b_2018-07-01.png (d300032c23fc...), FinalLightCurve_TrES-3 b_2018-07-01.csv (026b86e76c98...)

Compute resources

Wall time 105.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-19 13:55Z.